

Linyi (Tiger) Hou

✉ linyh2@illinois.edu | github.com/TigerHou2 | Permanent Resident

EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign Jun 2021 - present
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign Aug 2017 - May 2021

SKILLS

Programming Languages C/C++, Python, MATLAB, Javascript
Libraries/Frameworks astropy | scipy | pybind11, Eigen | OpenMP, Simulink, NodeJS
Tools/Databases Git, GitHub, Slurm, MongoDB, ADQL

RELEVANT EXPERIENCE

University of Illinois Urbana-Champaign

Graduate Researcher Jun 2021 - present

- ❖ Developed, implemented, and assessed several navigation and attitude determination algorithms in C++ and Python.
- ❖ Collaborated with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483) Jun 2021 - May 2024

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern May 2019 - Aug 2019

- ❖ Automated weekly builds and integration tests in TortoiseSVN using Python.
- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.

HONORS AND AWARDS

Yee Fellowship 2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship 2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference 2024

American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship 2020

American Institute of Aeronautics and Astronautics (AIAA)